

TEEMIL : Towards Educational MCQ Difficulty Estimation in Indic Languages

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Abstract

Difficulty estimation of multiple-choice questions (MCQs) is crucial for creating effective educational assessments, yet remains underexplored in Indic languages like Hindi and Kannada due to the lack of comprehensive datasets. This paper addresses this gap by introducing two datasets, TEEMIL-H and TEEMIL-K, containing 4689 and 4215 MCQs, respectively, with manually annotated difficulty labels. We benchmark these datasets using state-of-the-art multilingual models and conduct ablation studies to analyze the effect of context, the impact of options, and the presence of the *None of the Above* (NOTA) option on difficulty estimation. Our findings establish baselines for difficulty estimation in Hindi and Kannada, offering valuable insights into improving model performance and guiding future research in MCQ difficulty estimation.

1 Introduction

Difficulty estimation of multiple-choice questions (MCQs) is a growing field in educational technology and natural language processing (NLP), driven by the demand for personalized learning and data-driven methods to assess student understanding and customize educational content (Sajja et al., 2023). Traditionally, educators manually create and calibrate MCQs, a process that is time-consuming and subjective, leading to interest in automated methods for generating MCQs with varying difficulty levels (Kurdi et al., 2019). Most research in MCQ difficulty estimation focuses on language learning through lexico-semantics or end-to-end modeling. While these approaches show potential, they are often limited by the diversity of MCQ formats across subjects. Additionally, the majority of studies are centered on English-language datasets

(Yaneva et al., 2024; Veeramani et al., 2024), leaving a significant gap in resources and methodologies for other languages.

For Indic languages like Hindi and Kannada, this gap is particularly pronounced due to the scarcity of MCQ datasets (Doddapaneni et al., 2022), which often lack the depth and alignment with curricular needs necessary for formal educational assessments (Yaneva et al., 2024). To address these challenges, we aim to (i) create MCQ datasets in Hindi and Kannada, (ii) develop specialized datasets for MCQ difficulty estimation in these languages, and (iii) establish evaluation benchmarks for difficulty estimation in Indic contexts. These efforts will help bridge gaps in Indic educational assessments and lay the foundation for future research in automated difficulty estimation across diverse linguistic backgrounds.

First, building on the work of Maity et al. (2024b), we created an educational MCQ dataset using high school textbooks in Hindi and Kannada, covering subjects such as history, sociology, geography, economics and physical education. The dataset includes 4215 MCQs in Kannada and 4689 MCQs in Hindi, complete with distractors and multiple-choice options, including *None of the Above* (NOTA).

Following this, each MCQ was annotated for difficulty levels similar to Liang et al. (2019), categorized as *easy*, *medium*, or *hard*, to ensure accuracy and consistency. Finally, we benchmarked state-of-the-art language models, including mBERT (Devlin et al., 2019), XLM-R (Conneau et al., 2020), and IndicBERT (Kakwani et al., 2020), revealing that Kannada presents greater challenges than Hindi, establishing baselines for future studies on MCQ difficulty estimation. We also provide an analysis of the challenges encountered with these datasets to guide future research in Indic educational assessments. The dataset will be publicly released

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Context: तालगुंदा शिलालेख कर्नाटक में मिला पहला संस्कृत शिलालेख है। कर्नाटक में वस्तुशिल्प कला की नींव कदंब राजवंश ने डाली थी। उन्होंने बनवासी में अनेक मंदिर और विहारों का निर्माण कराया। कदंबों के काल में प्राथमिक और उच्च शिक्षण का विकास हुआ, और कई शिक्षण केंद्र स्थापित किए गए। इनमें अग्रहार, ब्रह्मपुरी और घटिक स्थान प्रमुख थे। उस समय के प्रसिद्ध अग्रहार तालगुंद और बल्लिगावि थे। अग्रहार गुरुकुल के रूप में कार्यरत थे और आवासीय विद्यालय की तरह चलते थे।

Question: कर्नाटक में कदंब राजवंश द्वारा स्थापित शैक्षिक केंद्र का क्या नाम है?

Option A: अग्रहारा

Option B: ब्रह्मपुरी

Option C: घटक

Option D: अग्रावती

Difficulty Label: Medium

Figure 1: Samples from TEEMIL-H dataset.

to encourage further studies in Indic languages¹. Overall our contributions can be summarized as follows:

- We first create a dataset of MCQs in Hindi and Kannada from textbooks, covering a range of educational subjects.
- We develop TEEMIL-H and TEEMIL-K datasets with 4689 and 4215 MCQs for Hindi and Kannada respectively with annotated difficulty labels.
- We benchmark several state-of-the-art language models for difficulty estimation in Hindi and Kannada.
- We present ablation studies to identify various challenges in MCQ difficulty estimation for Hindi and Kannada.

The remainder of this paper is organized as follows: Section 2 reviews related work on MCQ datasets, generation techniques, and difficulty estimation methods. Section 3 details the construction and annotation of the dataset, followed by a comprehensive analysis in Section 4. Section 5 presents the experiments and results. Finally, Sections 6 and 7 conclude the paper with a discussion of limitations and implications for future research.

2 Related Work

In this section, we provide an overview of works related to existing MCQ datasets, automatic MCQ generation, and MCQ difficulty estimation, highlighting the key challenges and open issues in each area.

¹The full dataset is available in <https://github.com/manikandan-ravikiran/TEEMIL>.

MCQ Datasets The development of multiple-choice question (MCQ) datasets has been crucial for advancing difficulty estimation. Early datasets like SQuAD (Rajpurkar et al., 2016), HotpotQA (Yang et al., 2018), and Natural Questions (Kwiatkowski et al., 2019) focused primarily on non-educational domains such as Wikipedia, offering valuable resources for question-answering tasks but lacking the specific educational context needed for pedagogical assessments. Similarly, datasets like NewsQA (Trischler et al., 2017), which focus on narrative content, are not tailored for educational purposes. Many of these datasets emphasize commonsense inference and complex question answering (Dhingra et al., 2017; Zellers et al., 2018; Yu et al., 2020; Sil et al., 2023). Educational datasets like RACE (Lai et al., 2017) and CLOTH (Xie et al., 2018) have been developed to assess reading comprehension in standardized tests, providing insights into question difficulty. However, these datasets are primarily limited to English, highlighting a significant gap for other languages. Datasets such as Crowdsourced MCQs (Welbl et al., 2017), LearningQ (Chen et al., 2018), ARC (Clark et al., 2018), Textbook MCQ (Li et al., 2018) and EduQG (Hadifar et al., 2022) are derived from online courses and textbooks but predominantly focus on English, underscoring the need for more diverse linguistic resources. Efforts to address these gaps with MCQ datasets in Indic languages include the XOR-QA (Asai et al., 2021), which includes Telugu as part of its cross-lingual evaluation and Belebele dataset (Bandarkar et al., 2023) which contains questions in Hindi and Kannada. Despite their value, these datasets are small, limited in scope, and lack annotations for difficulty estimation.

MCQ Generation Methods for generating multiple-choice questions (MCQs) have significantly evolved from early ontology-based (Pa-

pasalouros et al., 2008) and dependency-based (Afzal and Mitkov, 2013) approaches to advanced machine learning and transformer-based models. While these early methods were effective in controlled environments, they lacked adaptability across various subjects and contexts (Mitkov et al., 2006). The advent of neural networks and advanced transformer models (Correia et al., 2012; Vachev et al., 2022) has led to substantial improvements in generating contextually relevant and grammatically correct questions, especially when models are fine-tuned on domain-specific data (Jahangir et al., 2024). More recently, various methods have been explored, including integrating masked language models to enhance semantic understanding and syntactic accuracy (Matsumori et al., 2022), fine-tuning models like T5 on educational datasets (Wang et al., 2023), employing hybrid methods (Kumar et al., 2023) that combine ontologies with machine learning for automatic MCQ generation, prompt-based methods (Kalpakchi and Boye, 2023; Kiyak and Kononowicz, 2024) and multi-step large language models (Maity et al., 2024a; Xiong et al., 2022). However, these methods have primarily been evaluated for English and lack adaptability for Indic languages.

MCQ Difficulty Estimation The task of predicting the difficulty of multiple-choice questions (MCQs) has evolved from traditional feature-based approaches (Freedle and Kostin, 1993; Perkins et al., 1995), which were less effective for domain-specific problems (Yasmine H. El Masri and Baird, 2017) due to their inability to adequately capture the complexities of varied content (Beinborn et al., 2014; Susanti et al., 2017), to neural network models that leverage advanced techniques from various fields (Huang et al., 2017; Hsu et al., 2018). The introduction of transformer models has further improved accuracy in difficulty prediction across multiple subjects, including programming, mathematics, and computer science (Zhou and Tao, 2020; Benedetto et al., 2020). Advancements in pretraining techniques have significantly enhanced difficulty estimation in math and computer science questions (Loginova et al., 2021), and large language models (LLMs) have also been employed to refine difficulty prediction for USMLE medical MCQs (Veeramani et al., 2024; Ram et al., 2024; Dueñas et al., 2024; Rogoz and Ionescu, 2024; Gombert et al., 2024). Beyond difficulty prediction, some models have been applied to generate questions at specific difficulty levels (He et al., 2021;

Liang et al., 2019). However, much of this research has focused on narrow domains and predominantly uses English as the language of study.

3 Dataset Construction

In this section, we describe the data construction process for the TEEMIL-K and TEEMIL-H datasets. The construction of these datasets was a collaborative effort that involved a team of two school instructors, two natural language processing experts, and four students from classes 8 to 11 (Appendix C).

3.1 Data Source

The primary challenge in developing educational applications for Indic languages is the scarcity of openly accessible resources. While global initiatives have promoted Open Educational Resources (OER), most efforts are focused on English, leaving Indic languages underserved. Moreover, many Indian textbooks are protected by strict copyright laws, restricting their use in educational technologies.

To address this, we first sourced textbooks for classes 6 to 12 from the Karnataka Text Book Society (KTBS)². These textbooks, provided in epub format under a permissive license, were selected for their adaptability and suitability for integration into educational tools.

The EPUB files were converted into plain text (TXT) format to enable more efficient processing. Irrelevant content, such as prefaces and administrative details, was manually removed to retain only the core instructional material. For the Hindi and Kannada datasets, a total of 22 textbooks covering subjects like history, civics, geography, economics, and physical education were selected. However, due to formatting inconsistencies, the physical education textbooks for Hindi were excluded from further processing. From the remaining textbooks, approximately 15,000 paragraphs were extracted for both Kannada and Hindi. Subsequently, two subject-matter instructors curated around 5,000 key paragraphs per language for MCQ generation, adhering to the guidelines outlined in Appendix A.

²The Karnataka Textbook Society (KTBS) provides textbooks in multiple Indic languages through its website <https://www.ktbs.kar.nic.in/>. However, the majority of available textbooks are in Kannada and Hindi.

3.2 MCQ Creation

Due to the lack of openly available educational MCQ datasets in Hindi and Kannada, and to address the inefficiencies of manual MCQ creation, we adapted an Automated MCQ Generation framework using Multistage Prompting (MSP) (Maity et al., 2024b). The original MSP framework involves four sequential stages: (a) paraphrasing, (b) key phrase identification, (c) question generation, and (d) distractor generation. However, MSP is resource-intensive, primarily because it requires separate prompts for each stage and relies on large models like GPT-4 (OpenAI, 2023). Besides, initial testing of MSP on Hindi and Kannada paragraphs showed that the paraphrasing stage offered minimal improvement in question diversity, and the distractor generation yielded similar results to simpler single-stage approaches.

To improve the efficiency of the process, we adapted the MSP framework by (i) removing the paraphrasing and distractor generation stages, (ii) merging key phrase identification and question generation into a single prompt, and (iii) replacing GPT-4 with the LLaMA-3-70B model (Touvron et al., 2023), which is pre-trained on 30 non-English languages, including Hindi and Kannada. The updated prompt instruction is: *For the input paragraph <paragraph>, first identify the key phrases <keyphrase> and using them create five multiple-choice questions with answers in the original language.* This simplified approach improves the efficiency of MCQ generation while maintaining relevance and accuracy. Using this adapted framework, each paragraph generated five MCQs, producing approximately 25,000 MCQs for both Hindi and Kannada.

	TEEMIL-H	TEEMIL-K
MCQ's	4689	4215
Number of NOTA	487	132
Avg. length of answers ^{α}	18	28
Avg length of questions ^{α}	55	54
Avg length of paragraphs ^{α}	740	593
Remember ^{β}	2850	2706
Understand ^{β}	1819	1463
Apply ^{β}	11	38
Analysis ^{β}	10	2

Table 1: Dataset statistics of TEEMIL-H and TEEMIL-K dataset. α : Number of characters, β : Bloom taxonomy statistics.

3.3 MCQ Annotation for Difficulty

The annotation process for our dataset was done in stages, each of which is described below:

MCQ Sampling: The two instructors were tasked with selecting one key MCQ per paragraph. The MCQs were chosen based on several criteria: grammatical clarity, answerability, diversity, complexity, and alignment with Bloom’s Taxonomy levels (Appendix B). This process resulted in 4689 MCQs for Hindi and 4215 MCQs for Kannada. Detailed statistics for this sampling are provided in Table 1. Also the final MCQ’s consists of all the selected subjects.

Annotator Training: The student annotators underwent comprehensive training sessions on how to annotate the MCQs through google sheet (Appendix C). Each student annotator was asked to solve the MCQs and rate the difficulty as *easy*, *medium*, or *hard* based on their inherent understanding. This method aimed to capture the natural perception of difficulty from a student’s perspective. This process continued until each MCQ was solved by at least two annotators.

Inter-Annotator Agreement (IAA): To evaluate the consistency of the annotations, an Inter-Annotator Agreement study was conducted, where each sample was annotated by two annotators. Cohen’s kappa (Fleiss and Cohen, 1973) was calculated for each language to measure the level of agreement between the annotators. The kappa score was 0.65 for Hindi and 0.69 for Kannada, indicating a substantial agreement between the annotators.

Final Annotation: After achieving satisfactory IAA scores, the NLP researchers proceeded with the final annotation of the dataset. If two annotators assigned the same label to an MCQ, that label was used. In cases of disagreement, a follow-up questionnaire (Appendix D) was discussed with the student annotators. The questionnaire included targeted queries to help determine the final difficulty annotation. Based on the responses, the NLP researchers assigned the final labels, resulting in the finalized TEEMIL-H and TEEMIL-K datasets. Example MCQ with difficulty labels are as shown in Figure 1.

4 Data Analysis

In this section, we analyze the properties of the TEEMIL-H and TEEMIL-K datasets. Since the estimation of difficulty depends on factors such as the options, distractors, and context, we present three types of analysis: question analysis, option-label analysis, and Bloom’s taxonomy analysis. Mean-

while, Table 2 compares TEEMIL with other educational MCQ datasets.

4.1 Question Analysis

We analyzed the distribution of question types in the TEEMIL-H and TEEMIL-K datasets, each consisting of multiple-choice questions (MCQs) generated using a large language model (LLM). The average question length is approximately 55 characters. To understand the relationship between question types and difficulty, we employed a heuristic method that categorized questions based on the presence of common interrogative words such as What, Who, How, When, Where, Which and Why. The resulting distribution is detailed in Appendix E.

Upon reviewing the distribution data, clear differences emerge between the TEEMIL-H and TEEMIL-K datasets. The TEEMIL-H dataset, which focuses on subjects like history, civics, geography, and economics, shows a predominance of certain question types. Specifically, the What (2078 instances), Which (1344 instances), and Who (678 instances) categories are heavily represented. This suggests that the Hindi curriculum leans toward questions that focus on factual recall, which may skew the overall difficulty distribution. Notably, there is a significant presence of How questions (348 instances), which are typically associated with more complex cognitive demands, but they are still fewer in number compared to the fact-based questions.

On the other hand, the TEEMIL-K dataset, which encompasses a broader curriculum that includes subjects like physical education along with history and civics, shows a different pattern. The What category is even more dominant here, with 3173 instances. However, the Which (330 instances) and Who (416 instances) categories are significantly less frequent than in TEEMIL-H, reflecting a different focus in question formulation. The How category, which could suggest more open-ended or higher-order thinking questions, is notably underrepresented in TEEMIL-K, with only 8 instances, indicating a potential focus on simpler or more direct questions. The Other category, with 188 instances, highlights the broader variety of questions present in the Kannada dataset.

These differences in question distribution point to variations in the cognitive demands of the two curricula. The TEEMIL-H dataset’s emphasis on fact-based questions (What, Which, and Who) likely skews the difficulty spectrum towards either easy

or hard questions, with fewer medium-difficulty questions. In contrast, the TEEMIL-K dataset shows a broader mix of question types and, thus, a more balanced distribution of difficulty levels.

4.2 Option Analysis

Following the methodology of [Rodriguez-Torrealba et al. \(2022\)](#), we evaluated the quality of generated options using several automated metrics, including BLEU (1 to 4-grams) ([Papineni et al., 2002](#)), ROUGE-L ([Lin, 2004](#)), and cosine similarity ([Buck and Koehn, 2016](#)). For each sample, scores were calculated by comparing the correct answer with three generated distractor options. This evaluation was repeated for all the MCQ’s and the average scores for each language are shown in Table 3. Subsequently, we analyzed the relationship between BLEU scores, question types, and their ground truth difficulty labels.

The BLEU scores across both languages revealed a consistent pattern, with TEEMIL-K MCQs achieving higher BLEU scores than TEEMIL-H MCQs. Specifically, TEEMIL-K BLEU-1 score was 0.44115, compared to 0.38785 for TEEMIL-H, and this trend persisted across all n-gram levels. This suggests that the distractors in TEEMIL-K are more lexically similar to the correct answers than those in Hindi. Moreover, TEEMIL-K also exhibited higher cosine similarity values (0.1467) compared to TEEMIL-H (0.11975), indicating a greater contextual similarity between distractors and correct answers in TEEMIL-K.

The granular analysis of BLEU scores across question types highlights clear trends in the relationship between distractor quality and question difficulty. Specifically, questions categorized as How consistently exhibit the lowest BLEU scores, averaging 0.389. This low similarity between distractors and correct answers suggests that these questions are inherently more challenging. Conversely, What, Who, and When questions, which are generally more fact-based, show moderate BLEU scores of around 0.41. This intermediate score corresponds to a more balanced distribution of difficulty labels, with these questions spanning easy, medium, and hard difficulty levels. This pattern suggests that the distractors for these questions are more moderately challenging and not drastically different from the correct answers, resulting in a more even spread of difficulty levels. Finally, the Other category of questions achieves the highest BLEU scores, averaging 0.43. The higher similar-

Dataset	Size	Type	Data	Domain	Difficulty Estimation
LearningQ (Chen et al., 2018)	230K	E	document	Online courses	No
OpenBookQA (Banerjee et al., 2019)	6K	M	paragraph	textbook	
TQA (Li et al., 2018)	26K	M	document	textbook	
SciQ (Welbl et al., 2017)	13K	M	paragraph	textbook	
ProcessBank (Berant et al., 2014)	0.5k	M	paragraph	textbook	
EduQG (Hadifar et al., 2022)	3K	M	document	textbook	
Social Questions (Scaria et al., 2024)	0.6K	E	document	textbook	
Turkish QG (Çagatay Akyön et al., 2021)	4K	M	paragraph	textbook	
MedicalMCQ (Yaneva et al., 2024)	0.9K	M	paragraph	Standardized tests	Yes
TEEMIL (Ours)	≈8K	M	paragraph	textbook	

Table 2: Qualitative comparison of educational MCQ datasets for difficulty estimation. E: Extractive questions, M: Multiple-choice questions.

Dataset	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE-L	CS
TEEMIL-H	38.78	27.49	22.54	19.11	18.61	11.97
TEEMIL-K	44.11	31.68	26.00	22.25	14.94	14.67

Table 3: Evaluation of distractor quality in the provided options. All values are reported as percentages.

ity between distractors and correct answers implies that these questions tend to be easier, as reflected by the assignment of a higher proportion of *easy* labels. The closer alignment between distractors and the correct options likely reduces the cognitive load required to eliminate incorrect answers, thus simplifying the overall difficulty of these questions.

4.3 Bloom Taxonomy Analysis

Bloom’s Taxonomy, which classifies cognitive skills from basic recall to complex analysis, offers a useful framework for estimating MCQ difficulty. Assigning Bloom’s levels to each question helps establish a link between cognitive demand and difficulty. In our dataset, approximately 60% of the questions fall under the Remember category of Bloom’s Taxonomy (see Table 1), with the remaining questions distributed across higher cognitive levels. This distribution is consistent with findings in educational research, which suggest that around 70% of classroom questions target lower cognitive skills, such as recall, while the remaining focus on higher-order thinking skills, including application and analysis (Tofade et al., 2013). Structuring the dataset to align with these cognitive levels increases its efficacy in training models to predict MCQ difficulty. This also ensures that models are exposed to a diverse range of cognitive challenges, accurately representing the spectrum of question difficulty typically found in educational settings.

5 Experiments

In this section, we detail the different models, describe their experimental configurations, and

present the results obtained for the TEEMIL-H and TEEMIL-K datasets, respectively.

5.1 Models and Experimental setup

In this study, we formalize difficulty estimation as a classification task and use the architecture proposed by Raina and Gales (2022) for MCQ complexity prediction. For our experiments we employ a transformer-based architecture with models such as mBERT, XLM-R, and IndicBERT, each optimized for processing Indic languages.

The input format for each multiple-choice question (MCQ) follows the structure: [CLS] Context [SEP] Question [SEP] Option A [SEP] Option B [SEP] Option C [SEP] Option D, where the context precedes the question and its four answer options, each delimited by the [SEP] token. This sequence is fed into the transformer encoder to produce contextualized representations, which are subsequently processed by a classification head. The classification head outputs a probability distribution over the difficulty levels (i.e., easy, medium, hard). During inference, the model assigns the difficulty label corresponding to the option with the highest predicted probability.

All models are fine-tuned on their pretrained weights, following the experimental protocol outlined in Appendix F. The dataset is split into training and testing sets in an 80:20 ratio. Post-training, each model is evaluated using standard performance metrics Precision (P), Recall (R), and F1-Score (F1) to assess its accuracy in predicting difficulty levels.

5.2 Results and Analysis

In this section, we establish baseline results for difficulty estimation by investigating key factors such as input context, options and presence of NOTA.

Baseline Results: The evaluation of multiple transformer-based models mBERT, XLM-R, and IndicBERT on the TEEMIL-K (Kannada) and TEEMIL-H (Hindi) datasets offers valuable insights into their effectiveness in difficulty estimation (Table 4). Among the models, XLM-R demonstrates superior performance, achieving the highest F1-scores of 0.9681 for Hindi and 0.8987 for TEEMIL-K, highlighting its robustness in handling diverse linguistic structures. While mBERT performs well, particularly on Hindi (F1: 0.9247), it lags behind XLM-R in TEEMIL-K. IndicBERT, on the other hand, shows lower performance, with F1-scores of 0.6887 for TEEMIL-K and 0.5415 for TEEMIL-H, indicating limitations in its handling of these tasks.

	TEEMIL-H			TEEMIL-K		
	P	R	F1	P	R	F1
mBERT	0.9555	0.9045	0.9247	0.8757	0.8644	0.8597
XLM-R	0.9814	0.9457	0.9681	0.8965	0.9010	0.8987
IndicBERT	0.4873	0.6528	0.5415	0.7015	0.7193	0.6887

Table 4: Results of difficulty estimation using transformer models.

Further analysis, using BLEU scores (see Table 3), reveals that TEEMIL-K produces consistently higher BLEU-1 scores compared to TEEMIL-H, with values of 44.115 for TEEMIL-K and 38.785 for TEEMIL-H. The higher BLEU scores for TEEMIL-K indicate greater lexical similarity between distractors and correct answers, likely making it harder for models to differentiate difficulty levels. This is further supported by higher ROUGE-L and cosine similarity scores for TEEMIL-K, suggesting stronger contextual alignment between distractors and correct answers. Consequently, XLM-R and mBERT exhibit stronger performance on TEEMIL-K, likely due to their ability to handle the subtle difficulty levels associated with more closely related distractors.

An in-depth examination of the confusion matrices for XLM-R (see Appendix H) provides additional insights. For TEEMIL-H, errors are more concentrated, suggesting greater stability across difficulty levels, with most misclassifications occurring between medium and other two categories. For TEEMIL-K, the errors are more dispersed, implying challenges in consistently classifying difficulty levels. The models struggle to distinguish between *easy*, *medium*, and *hard* categories, particularly in medium and hard questions. The difficulty in differentiating these categories may stem from subtle and context-dependent factors, such as the presence of strong distractors or ambiguities in the

labeling process, influenced by factors like student variability and cognitive abilities. Lastly we can see IndicBERT to underperform significantly, we think this has to do with the pretraining data and model simplicity of IndicBERT, however more experiments are needed in this regard.

Effect of Options: To assess the impact of options on difficulty estimation, we trained the transformer models using only the [CLS] Context [SEP] Question [SEP] input format, excluding the options and distractors. The results, presented in Table 5, demonstrate that the inclusion of options significantly influences model performance. When options are present, the models gain additional contextual clues from the distractors. However, when the distractors are too similar to the correct answer, as indicated by higher BLEU scores, model performance declines due to the increased difficulty of distinguishing between the correct answer and the distractors. On the other hand, when options are excluded, the task becomes simpler, leading to improved model performance, particularly in cases where the distractors closely resemble the correct answer. These findings suggest that while options enhance the context for the model, their distinctiveness and quality play a crucial role in determining overall model accuracy in difficulty estimation tasks.

	TEEMIL-H			TEEMIL-K		
	P	R	F1	P	R	F1
mBERT	0.9422	0.9457	0.943	0.8885	0.8903	0.8894
XLM-R	0.9798	0.9611	0.9698	0.9071	0.9086	0.9077
IndicBERT	0.7086	0.7571	0.7192	0.8118	0.8112	0.8855

Table 5: Results of difficulty estimation using transformer models with the exclusion of options from the input.

Effect of Context: In our earlier results (Table 4), we reported a high F1 score when the model was provided with the context (paragraph), MCQ, and options. Interestingly, we observed that the F1 score increased further upon the removal of the options. This prompted a deeper investigation into the role of context in difficulty estimation.

To isolate the effect of context, we conducted an ablation study by removing the paragraph (context) and using only the MCQ and options as inputs. The results of this experiment, summarized in Table 6, show an improvement in the model’s performance for TEEMIL-H, surpassing both the original baseline and the variant without options. Meanwhile for TEEMIL-K, there is a drop in the results.

This unexpected improvement for TEEMIL-H may be attributed to two factors: (a) The model processes all inputs simultaneously, rather than evaluating each element (context, MCQ, and options) separately for its contribution to the difficulty. As a result, it may rely on simpler patterns in the question phrasing instead of fully leveraging the contextual information (McCoy et al., 2019). Removing the context simplifies the task, allowing the model to perform better by focusing on more straightforward elements. (b) The annotators (students), being non-native speakers, may have mentally translated or interpreted the questions into their native language when assigning difficulty labels. Although the model does not receive these translations directly, the annotators’ simplified understanding of the questions could have influenced the labels (Pavlick and Kwiatkowski, 2019). This inherently leads to proxy patterns, which are more easily picked up by the model, as these patterns are simpler and align with the annotators’ interpretation rather than the full linguistic and contextual complexity. As a result, performance scores improve and lead to skewed difficulty estimations.

	TEEMIL-H			TEEMIL-K		
	P	R	F1	P	R	F1
mBERT	0.9742	0.9852	0.9787	0.6086	0.6258	0.5741
XLM-R	0.9825	0.9864	0.9843	0.7122	0.6135	0.5483
IndicBERT	0.9329	0.8874	0.9060	0.4679	0.5874	0.5186

Table 6: Results of difficulty estimation using transformer models with the exclusion of context from the input.

Effect of NOTA: To assess the generalization capacity of the models, we adopt the approach outlined by Wang et al. (2024), introducing *None of the Above* (NOTA) as an answer option in the test sets for both TEEMIL-H and TEEMIL-K, while keeping the training set unchanged. This adjustment compels the models to identify patterns to determine whether all given options are incorrect. The results, presented in Table 7, indicate a slight decline in performance across both datasets with the introduction of NOTA. For example, in TEEMIL-K, the F1-score for XLM-R decreases from 0.8987 (without NOTA) to 0.859 (with NOTA), with a similar trend observed in TEEMIL-H. We hypothesize that the inclusion of NOTA introduces ambiguity, thereby increasing the complexity of the task. For easier questions, an incorrect selection of NOTA may lead to an overestimation of difficulty, while for more challenging questions, the presence of strong dis-

tractors may cause the model to incorrectly select NOTA. These results suggest that the addition of NOTA poses additional challenges for difficulty estimation, complicating the models’ ability to accurately classify question difficulty.

	TEEMIL-H			TEEMIL-K		
	P	R	F1	P	R	F1
mBERT	0.9315	0.9366	0.9330	0.8630	0.8279	0.8388
XLM-R	0.9316	0.9406	0.9359	0.8634	0.8566	0.8590
IndicBERT	0.5312	0.4932	0.4903	0.5895	0.5813	0.5725

Table 7: Results of difficulty estimation with the substitution of *None of the Above* (NOTA) in the options.

6 Conclusion

In this work, we introduce the task of multiple-choice question (MCQ) difficulty estimation for Indic languages, focusing on Kannada and Hindi, through the development of two comprehensive datasets: TEEMIL-K and TEEMIL-H. Each dataset contains over 4,000 manually annotated MCQs, designed to support benchmarking in this domain. We evaluate state-of-the-art transformer models, including mBERT, XLM-R, and IndicBERT, to establish baselines for difficulty estimation. Our experiments reveal that XLM-R consistently outperforms the other models on both datasets, setting new benchmarks, while IndicBERT demonstrates significant limitations in handling linguistic diversity across these languages.

Our study also highlights the effects of options and context on the performance of difficulty estimation, revealing several side effects. Additionally, we assess the impact of introducing the *None of the Above* (NOTA) option during testing, which adds complexity and results in a noticeable decline in model performance. This underscores the challenge of estimating question difficulty when ambiguous options like NOTA are present.

Overall, we believe the TEEMIL datasets, along with the benchmarks established in this work, will be valuable resources for future research, contributing to the development of automated educational tools tailored to Indic languages.

7 Limitation

This research has several limitations that leave room for further exploration. First, the study was restricted to two Indic languages, Kannada and Hindi, due to the lack of open corpora and the high cost of annotation for other low-resource languages, limiting the broader applicability of the

findings. Second, the dataset used was focused only on subjects history, civics, geography, economics, and physical education resulting in limited coverage. Third, the difficulty estimation relied on only three transformer models (mBERT, XLM-R, and IndicBERT), without considering the performance of smaller or simpler models, which may have provided additional insights. Finally, the introduction of the *None of the Above* (NOTA) option in MCQs posed a significant challenge. Although NOTA adds complexity to difficulty estimation, its impact was not explored in depth. Addressing how models handle the ambiguity introduced by NOTA is crucial for accurately estimating question difficulty and remains an open area for future research.

Ethics Statement

The data collection process for this study did not involve direct interaction with human participants that would require formal ethics board approval. The datasets utilized are publicly available and licensed for redistribution (See Appendix I). Annotations for difficulty levels were provided by students. We paid participants at a rate that was above both the living wage in our jurisdiction and Prolific’s current recommendation of at least 9.00 GBP/12.00 USD (Shmueli et al., 2021). By involving students in the annotation process, we ensured that the dataset reflects authentic insights into the perceived difficulty of questions, aligning more closely with real-world educational contexts.

We believe that the NLP systems developed from these datasets have the potential to greatly enhance educational assessment tools, fostering more effective and inclusive learning environments. When applied appropriately, these models and datasets can help deliver personalized and equitable educational experiences, improving student learning outcomes across diverse backgrounds.

However, we acknowledge potential biases in both the datasets and the models. While the datasets are constrained to specific languages and subjects, we have included educational content spanning classes 6 to 12 to mitigate bias toward any particular group of students. Moving forward, we aim to expand the scope of the data to include more languages and educational domains, ensuring greater inclusivity and fairness in the development of educational NLP systems.

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A Guidelines for Paragraph Selection

The guidelines used for paragraph selection is as follows

- **Focus on Learning Goals:** Choose paragraphs that cover important concepts and skills the students need to learn.
- **Bloom’s Taxonomy:** Select paragraphs for different question types:
 - Easy: Basic facts and definitions.
 - Medium: Paragraphs that explain or apply concepts.
 - Hard: Complex ideas that require analysis or evaluation.
- **Balance Cognitive Load:** Pick paragraphs that range from simple recall to more complex thinking:
 - Easy: Straightforward information.
 - Medium: Multiple ideas but manageable.
 - Hard: Deep understanding and critical thinking.
- **Transfer of Learning:** Ensure some paragraphs help students apply knowledge to new situations:
 - Easy: Directly related to what they’ve learned.
 - Medium: Slightly different context.
 - Hard: Completely new situations.
- **Contextual Complexity:** Choose paragraphs based on how clear or complex they are:
 - Easy: Simple, direct information.
 - Medium: Some complexity but clear.
 - Hard: Requires understanding complex ideas.
- **Topic Coverage:** Select paragraphs from a variety of topics to cover the entire subject area.

- **Relevance to Core Ideas:** Ensure the paragraphs are directly related to the main ideas students need to know.
- **Length of Paragraphs:** Choose paragraphs that are the right length for creating good questions, not too long or too short.
- **Examples and Case Studies:** Include paragraphs with examples or cases where students can apply their knowledge.
- **Mix of Difficulty:** After selection, make sure there’s a balance of easy, medium, and hard paragraphs to create a range of question difficulties.

B MCQ selection Criteria

- **Alignment with Bloom’s Taxonomy:** Ensure that the MCQ corresponds to one of the cognitive levels in Bloom’s Taxonomy: Remember, Understand, Apply, Analyze, Evaluate, or Create. Each question should engage the student with an appropriate level of cognitive demand based on the content of the paragraph.
- **Grammatical Clarity:** The MCQ must be grammatically correct and easy to understand, ensuring clarity in both the question and answer options.
- **Answerability:** The question must have a clear, unambiguous answer based on the information provided in the paragraph. Avoid questions where the correct answer is not directly supported by the text.
- **Diversity:** Ensure that there is a diversity of question types (e.g., what, who, how, etc.) across the selected MCQs to cover different forms of inquiry. Avoid repetition of similar question styles within the dataset.
- **Complexity:** Consider the complexity of the question. It should challenge students at an appropriate level, based on the content of the paragraph and Bloom’s level.

C Annotation on Google Sheets

Annotation done using google sheets via picker as shown in Figure 2. The annotation team details are in Table 8. All the instructors are school teachers with Masters Degree, teaching the subjects which are selected for MCQ generation.

Paragraph	MCQ	Option 1	Option 2	Option 3	Option 4	Difficulty
ಕೃಷಿಯಲ್ಲಿ ಯಾಂತ್ರಿಕರಣದ ಅಳವಡಿಕೆಯಿಂದ ಟ್ರ್ಯಾಕ್ಟರ್ ಉತ್ಪಾದಿಸುವ ಘಟಕಗಳು, ರಾಸಾಯನಿಕ ಗೊಬ್ಬರಗಳ ಕಾರ್ಖಾನೆ, ಕೃಷಿನಾಣಕಗಳ ಕಾರ್ಖಾನೆಗಳು ಹಾಗೂ ವಿವಿಧ ಹೈಬ್ರಿಡ್ ಬೀಜ ತಯಾರಿಸುವ ಕಂಪನಿಗಳು ವಿಸ್ತೃತಗೊಂಡಿವೆ ಹಸಿರು ಕ್ರಾಂತಿಯ ದುಷ್ಪರಿಣಾಮಗಳಾಗುವಂತೆ ಪ್ರದೇಶಗಳ ಸಂರಕ್ಷಣೆಯನ್ನು ಬದಲಿಸಿತು ಆದರೆ ಹಸಿರು ಕ್ರಾಂತಿಯು ದುಷ್ಪರಿಣಾಮಗಳನ್ನು ಕೂಡ ಬೀರಿತು ಅವುಗಳಲ್ಲಿ ಪ್ರಮುಖವಾಗಿ 1 ವೈಯಕ್ತಿಕ ಅಸಮಾನತೆ ಹಸಿರು ಕ್ರಾಂತಿಯು ಗ್ರಾಮೀಣ ವಲಯದ ಅರ್ಥಿಕತೆಯಲ್ಲಿ ಎರಡು ಬಗೆಯ ರೈತ ವರ್ಗವನ್ನು ಸೃಷ್ಟಿಸಿ ಅವರ ನಡುವೆ ಆದಾಯ ಅಸಮಾನತೆಯ ಕಂಡಕವನ್ನು ಹೆಚ್ಚು ಮಾಡಿದೆ ಅವರಿಂದ 1 ಶೇಮಂಡರ ರೈತರು-ಸಾಮಾನ್ಯ ಮಧ್ಯಮ ಹಾಗೂ ದೊಡ್ಡಭೂ ಹಿಂಡುವಳಿದಾರರು 2 ಬಡರೈತರು ಹಾಗೂ ಕೃಷಿ ಕಾರ್ಮಿಕರು-ಸಾಮಾನ್ಯವಾಗಿ ಅತಿ ಸಣ್ಣ ಹಾಗೂ ಸಣ್ಣ ಭೂ ಹಿಡುವಳಿದಾರರು ಹಾಗೂ ಕೃಷಿ ಕಾರ್ಮಿಕರು ಏಕೆಂದರೆ ಹಸಿರು ಕ್ರಾಂತಿಯ ಬಹುಪಾಲು ಪ್ರಯೋಜನ ಪಡೆದವರಿಂದ ದೊಡ್ಡ ಭೂ ಹಿಡುವಳಿದಾರರಾಗಿದ್ದಾರೆ ಕೃಷಿಯಲ್ಲಿ ಯಾಂತ್ರಿಕರಣಕ್ಕೆ ಬೇಕಾದ ಸೌಲಭ್ಯಗಳಾದ ಬೆರಗಾವರಿ ಸೌಲಭ್ಯ, ಕೃಷಿ ಹಣಕಾಸು ಮುಂತಾದವು ಸುಲಭವಾಗಿ ದೊಡ್ಡ ರೈತರಿಗೆ ಲಭ್ಯವಾಯಿತು.	ಕೃಷಿಯಲ್ಲಿ ಹಸಿರು ಕ್ರಾಂತಿಯ ಮುಖ್ಯ ಫಲಿತಾಂಶವೇನು?	ಹೆಚ್ಚಿದ ಬೆಳೆ ಇಳುವರಿ	ರಾಸಾಯನಿಕ ಗೊಬ್ಬರ ಬಳಕೆ ಕಡಿಮೆಯಾಗಿದೆ	ಸುಧಾರಿತ ನೀರಾವರಿ ಸೌಲಭ್ಯಗಳು	ಹೆಚ್ಚಿದ ಆರ್ಥಿಕ ಅಸಮಾನತೆ	Easy Medium Hard
ಕೃಷಿಯಲ್ಲಿ ಯಾಂತ್ರಿಕರಣದ ಅಳವಡಿಕೆಯಿಂದ ಟ್ರ್ಯಾಕ್ಟರ್ ಉತ್ಪಾದಿಸುವ ಘಟಕಗಳು, ರಾಸಾಯನಿಕ ಗೊಬ್ಬರಗಳ ಕಾರ್ಖಾನೆ, ಕೃಷಿನಾಣಕಗಳ ಕಾರ್ಖಾನೆಗಳು ಹಾಗೂ ವಿವಿಧ ಹೈಬ್ರಿಡ್ ಬೀಜ ತಯಾರಿಸುವ ಕಂಪನಿಗಳು ವಿಸ್ತೃತಗೊಂಡಿವೆ ಹಸಿರು ಕ್ರಾಂತಿಯ ದುಷ್ಪರಿಣಾಮಗಳಾಗುವಂತೆ ಪ್ರದೇಶಗಳ ಸಂರಕ್ಷಣೆಯನ್ನು ಬದಲಿಸಿತು ಆದರೆ ಹಸಿರು ಕ್ರಾಂತಿಯು ದುಷ್ಪರಿಣಾಮಗಳನ್ನು ಕೂಡ ಬೀರಿತು ಅವುಗಳಲ್ಲಿ ಪ್ರಮುಖವಾಗಿ 1 ವೈಯಕ್ತಿಕ ಅಸಮಾನತೆ ಹಸಿರು ಕ್ರಾಂತಿಯು ಗ್ರಾಮೀಣ ವಲಯದ ಅರ್ಥಿಕತೆಯಲ್ಲಿ ಎರಡು ಬಗೆಯ ರೈತ ವರ್ಗವನ್ನು ಸೃಷ್ಟಿಸಿ ಅವರ ನಡುವೆ ಆದಾಯ ಅಸಮಾನತೆಯ ಕಂಡಕವನ್ನು ಹೆಚ್ಚು ಮಾಡಿದೆ ಅವರಿಂದ 1 ಶೇಮಂಡರ ರೈತರು-ಸಾಮಾನ್ಯ ಮಧ್ಯಮ ಹಾಗೂ ದೊಡ್ಡಭೂ ಹಿಂಡುವಳಿದಾರರು 2 ಬಡರೈತರು ಹಾಗೂ ಕೃಷಿ ಕಾರ್ಮಿಕರು-ಸಾಮಾನ್ಯವಾಗಿ ಅತಿ ಸಣ್ಣ ಹಾಗೂ ಸಣ್ಣ ಭೂ ಹಿಡುವಳಿದಾರರು ಹಾಗೂ ಕೃಷಿ ಕಾರ್ಮಿಕರು ಏಕೆಂದರೆ ಹಸಿರು ಕ್ರಾಂತಿಯ ಬಹುಪಾಲು ಪ್ರಯೋಜನ ಪಡೆದವರಿಂದ ದೊಡ್ಡ ಭೂ ಹಿಡುವಳಿದಾರರಾಗಿದ್ದಾರೆ ಕೃಷಿಯಲ್ಲಿ ಯಾಂತ್ರಿಕರಣಕ್ಕೆ ಬೇಕಾದ ಸೌಲಭ್ಯಗಳಾದ ಬೆರಗಾವರಿ ಸೌಲಭ್ಯ, ಕೃಷಿ ಹಣಕಾಸು ಮುಂತಾದವು ಸುಲಭವಾಗಿ ದೊಡ್ಡ ರೈತರಿಗೆ ಲಭ್ಯವಾಯಿತು.	ಕೃಷಿಯಲ್ಲಿ ಯಾಂತ್ರಿಕರಣದ ಮುಖ್ಯ ಫಲಿತಾಂಶವೇನು?	ರೈತರು	ಸಣ್ಣ ಪ್ರಮಾಣದ ರೈತರು	ದೊಡ್ಡ ಪ್ರಮಾಣದ ರೈತರು	ಕೃಷಿ ಕಾರ್ಮಿಕರು	

Figure 2: Annotation using google sheets.

Participants	Gender	Education	Native Language	Language of Annotation
Instructor 1	Female	Masters Degree	Tamil	NA
Instructor 2	Female	Masters Degree	Kannada	
Expert 1	Male	Masters Degree	Tamil/Kannada	
Expert 2	Male	Bachelors Degree	Hindi	
Student 1	Male	class 8	Tamil	Hindi
Student 2	Female	class 10	Marathi	Hindi
Student 3	Male	class 10	Kannada	Kannada
Student 4	Female	class 11	Tulu	Kannada

Table 8: Qualitative comparison of Annotation team

Question Category	TEEMIL-H	TEEMIL-K
How	348	8
What	2078	3173
When	17	21
Where	45	58
Which	1344	330
Who	678	416
Why	136	21
Other	44	188

Table 9: Question types in TEEMIL-H and TEEMIL-K dataset.

D Questionnaire for Resolving annotation ambiguity

- Did you find the answer directly in the text, or did you have to think about it more?
- Did you have to compare multiple options that seemed similar?
- Was there anything in the question that confused you or made you think harder than usual?
- Did you need to use any specific strategies (like elimination or inference) to find the correct answer?
- Was the language or vocabulary used in the question difficult or unfamiliar?
- Did the structure or length of the question make it harder to understand?

E Question Types in TEEMIL

Table 9 shows statistics of various question types in TEEMIL-H and TEEMIL-K dataset.

F Hyperparameters

Various hyperparameters are as shown in Table 10.

Model Name	mBERT	XLM-R	IndicBert
API	bert-base-multilingual-cased	xlm-roberta-base	ai4bharat/indic-bert
Batch Size	32	16	32
Learning Rate	5e-5		
Epochs	10		
Warm Up Steps	500		
Optimizer	AdamW		

Table 10: Experimental Setup used for difficulty Estimation

G Dataset Analysis - Contd

Here we assume that length of the paragraph as the direct determinant of the difficulty. As such we analyze the relationship between the length of paragraph and difficulty labels to form set of rules. See Figure 3 and 4 respectively.

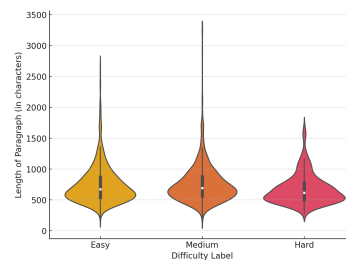


Figure 3: Relationship between length of paragraph and difficulty for TEEMIL-H.

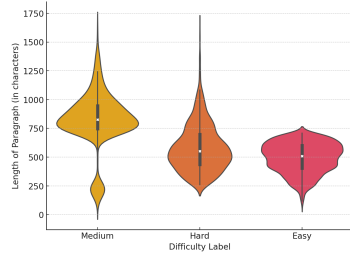


Figure 4: Relationship between length of paragraph and difficulty for TEEMIL-K.

H Confusion Matrices of XLM-R with TEEMIL

Confusion matrices are shown in Figure 5 & 6.

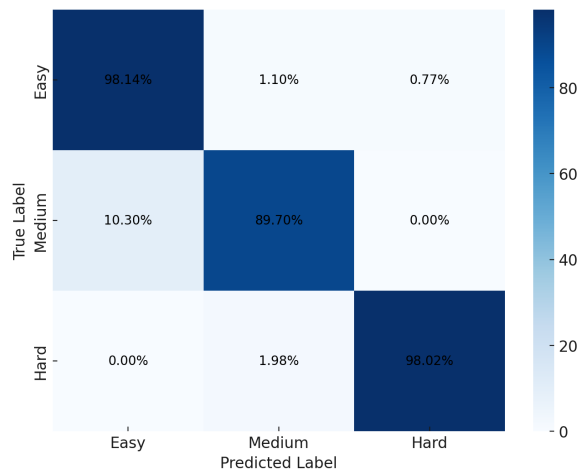


Figure 5: Confusion Matrix of XLM-R on TEEMIL-H.

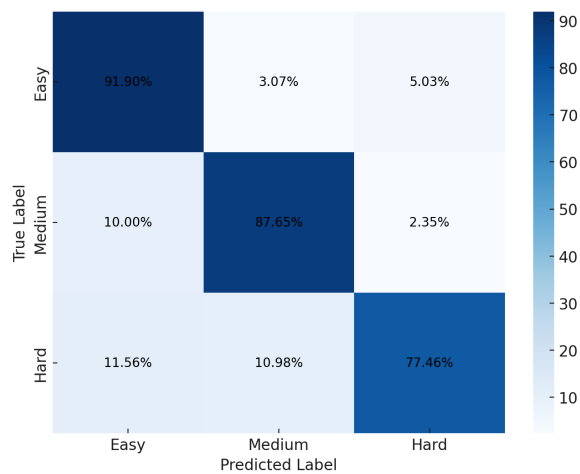


Figure 6: Confusion Matrix of XLM-R on TEEMIL-K.

I Copyright Details of Text book

Copyright detail is as in Figure 7.



Figure 7: Copyright and License details of Textbooks.